

Ali Awil

AI Researcher & Software Engineer

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EDUCATION

Master of Computer Science — Applied Artificial Intelligence Jan 2026 – Jun 2027

University of Ottawa

Machine Learning · Virtual Assistants · Deep Learning & Reinforcement Learning · Ethics in AI

BSc Computer Science (Co-op), Data Science Option — Summa Cum Laude Jan 2021 – Mar 2026

University of Ottawa

CGPA 9.32 / 10 · Dean's Honour List (multiple terms)

EXPERIENCE

AI & Demand Generation Engineering Intern May 2026 – Aug 2026

PureFacts, Toronto, ON (Hybrid; based in Ottawa, ON)

- Collaborate with data and AI engineers on ingestion, normalization, and warehouse-ready modeling sourced from SaaS CRM, collaboration, marketing, and external feeds supporting analytics and internal AI prototyping.
- Contribute to embedding and retrieval pipelines for unstructured corpora (meetings, documents): parsing, hybrid search tuning, and quality checks feeding responsibly grounded assistants.
- Leverage AI-assisted drafting where appropriate under senior review for specs and sprint planning; follow role-based access, privacy norms, and data-sensitivity constraints typical of regulated-client environments.

Teaching Assistant Jan 2026 – Apr 2026

University of Ottawa, Ottawa, Canada

- TA for CSI 3131 (Operating Systems), supporting students through labs and assignments in C and C++.
- Guided students through debugging in systems programming and reinforced core OS concepts including processes, threads, synchronization, and memory management.

Data Science Developer Intern May 2024 – May 2025

Department of National Defence, Ottawa, Canada

- Optimized model selection for FAM policy Q&A by benchmarking Longwizard-70B and Mistral Nemo, achieving **23% faster inference** and **89% accuracy**.
- Tested a fine-tuned LLaVA (WizardLM-70B + CLIP) model and analyzed outputs to evaluate multimodal reasoning quality.
- Led experimentation, benchmarking, and documentation for two large-scale initiatives (LLaVA multimodal and Space News).
- Validated entity-level accuracy and improved reliability of the Space News information extraction pipeline.

Software Engineer Intern May 2023 – Aug 2023

Zafin, Toronto, Canada

- Built an Enterprise RAG Knowledge Assistant using OpenAI's API achieving **90% accuracy** on internal benchmarks.
- Architected a custom RAG pipeline ingesting and processing 10K+ internal documents into a PostgreSQL vector database.
- Evaluated Falcon-7B vs GPT tradeoffs; deployed via Flask microservice on Azure.
- Co-authored "Closing the Gap Between Available Information and Accessible Knowledge," Zafin Engineering Blog (Medium, Aug 2023).

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, TypeScript, SQL, R

Frameworks & tools: PyTorch, Scikit-learn, Hugging Face, LangChain, FastAPI, Flask, React, Docker, Azure, Snowflake, Power BI, Git

Concepts: Machine learning, RAG, agentic AI, vector databases, NLP, deep learning, model evaluation, cloud deployment, data visualization

SELECTED PROJECTS

Voice Assistant Pipeline — ECAPA-TDNN verification, wake word, Whisper ASR, BERT intent/slots (20+ intents). huggingface.co/spaces/Aladenawil/atlas-voice-assistant

Flight Price Prediction — Random Forest, XGBoost, TensorFlow DNN on 300K+ records (DNN $R^2 \approx 0.97$); SHAP & paired t-tests. github.com/Aladenawil/flight-price-prediction

Food Prices BI Dashboard — Power BI trends dashboard for decision support. foodpricesdashboard.netlify.app

PedalPro Bike Service — Full-stack service booking application. pedalpro-bike-service.netlify.app

ShopSmart E-Commerce — E-commerce UI and flows. shopsmartecommerce.netlify.app

Simon Says Memory Game — Browser memory game. simonsaysgameuottawa.netlify.app